Deepa Krishnan

PhD Candidate, Materials Science
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Education

IIT Madras — Jul 2021 – Expected 2026
Ph.D., Metallurgical and Materials Engineering

Anna University — Aug 2019 – May 2021
M.E., Materials Science; CGPA 8.9/10

PSG College of Technology — Jul 2015 – May 2019
B.E., Mechanical Engineering; CGPA 8.2/10

Research and Teaching

Teaching Assistant, IIT Madras — Jan 2022 – May 2024
Chennai

- Ran laboratory sessions for the undergraduate course on phase transformations.

Visiting Researcher, Max Planck Institute for Iron Research — Jun 2023 – Sep 2023
Düsseldorf, Germany

- Characterised hydrogen embrittlement in high-entropy alloys with atom probe tomography.

Publications

1. Hydrogen trapping at grain boundaries in a CoCrFeNiMn high-entropy alloy. *Acta Materialia*, 2024.
2. Machine-learned interatomic potentials for refractory alloys. *npj Computational Materials*, 2023.
3. Corrosion of additively manufactured stainless steel in seawater. *Corrosion Science*, 2022.

Honours

- Prime Minister's Research Fellowship (2021)
- Best Oral Presentation, International Conference on Advanced Materials (2023)

Technical Skills

Characterisation: SEM, TEM, Atom Probe Tomography, XRD
Computation: Python, LAMMPS, VASP